

Ilyas Modni

PhD Candidate in Artificial Intelligence & Decision Tree Optimization

♥ Rabat, Morocco | 🇲🇦 Citizenship: Moroccan

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Research Interests

Machine Learning, Optimization, Decision Trees & Tree-based Methods, Explainable AI (XAI), Large Language Models (LLMs), Road Safety Analysis.

Education

- **Mohammadia School of Engineering (EMI)** Rabat, Morocco
Ph.D. Student in Artificial Intelligence & Decision Tree Optimization Since 2023
 - **Advisor:** Prof. Abdelaziz Berrado
 - **Thesis Focus:** Design and Development of an Intelligent System for Road Safety Data Valorization.
 - **Funding:** Recipient of the Excellence Grant (PASS - CNRST).
- **Faculty of Science Ben M'sik (FSBM)** Casablanca, Morocco
Master's in Data Science and Big Data (Graduated with Honors) 2023
- **Faculty of Science Ben M'sik (FSBM)** Casablanca, Morocco
Bachelor's in Mathematics and Informatics Sciences (Graduated with Honors) 2021

Experience

- **National Road Safety Agency (NARSA) Collaboration** Rabat, Morocco
PhD Researcher & AI Decision Support System Developer Since 2023
 - Developed a secure road safety data collection and monitoring platform aligned with the official pyramid.
 - Designed an AI agent for manipulating and communicating with tabular road safety data.
 - Implemented tree-based predictive models to estimate motorcycle accident severity.
 - Built an explainable rule-analysis module that identifies activated decision rules and key risk factors.
- **Manageering Inc.** Waterloo, Ontario (Remote)
Data Analyst & Market Research Team Member Nov. 2022 – Mar. 2025
 - Collaborated with a cross-functional team on a research project creating a Moroccan platform for smart home security.
 - Utilized BI & Data Analyst techniques to analyze the smart home security market, user behavior, and preferences.
- **Leoni Wiring Systems** Berrchid, Morocco
Data Scientist Intern Feb. 2023 – Aug. 2023
 - Designed and implemented dynamic KPI visualizations using BI tools and a standalone application to support real-time decision-making.
 - Applied advanced predictive algorithms to forecast KPI trends, enabling proactive business strategies.
 - Built a robust data pipeline to consolidate and centralize data for streamlined analysis.
- **Faculty of Science Ben M'sik (FSBM)** Casablanca, Morocco
Research Project Lead 2023
 - **Class Attendance Management System:** Created an innovative solution automating classroom attendance tracking utilizing facial recognition to identify and record student presence.

Publications

- [SITA 2025] I. Modni, et al. “Current Machine Learning Applications in Road Safety Monitoring: A Comprehensive Review.” *The 15th International Conference on Intelligent Systems: Theories and Applications*. DOI: 10.1109/SITA67914.2025.11273433. (Oct 2025).
- [IEOM 2025] I. Modni, et al. “An Overview of the Limitations of Traditional Greedy Tree-Based Methods.” *International Conference on Industrial Engineering and Operations Management*. Abstract & Presentation. DOI: 10.46254/af6.20250145. (Apr 2025).
- [GOL 2026] I. Modni, et al. “Towards Improved Decision Tree Interpretability via Large Language Models: A Case Study in Motorcycle Accident Severity.” *The 8th International Conference on Logistics Operations Management*. (Accepted for publication in Springer LNNS proceedings).
- [Engineering Applications of Artificial Intelligence] I. Modni, et al. “Actionable Predictive Analytics for Informed Road Safety Decisions: Case of Motorcycle Accidents in Morocco.” *Engineering Applications of Artificial Intelligence*. (Under Review).
- [Data Mining and Knowledge Discovery] I. Modni, et al. “A Survey of Tree-Based Methods: Taxonomy and Empirical Benchmark Analysis in Supervised Learning.” *Data Mining and Knowledge Discovery*. (Under Review).

Fellowships, Honors & Awards

- **PhD-Associate Scholarship (PASS) – Excellence Grant** Rabat, Morocco
Co-funded by the Ministry of Higher Education, Scientific Research and Innovation & CNRST Since 2023
- **Ma Thèse en 180 secondes (MT180)** Rabat, Morocco
University Finalist (Representative of EMI at Mohammed V University) May 2025

Selected Presentations

- **RESCIF Scientific Seminar** Rabat, Morocco
Presenter: “An Empirical Benchmark of Decision Tree Induction” (EMI) Apr 2026
- **AI For Road Safety Workshop** Rabat, Morocco
Presenter: “Actionable Predictive Analytics for Informed Road Safety Decisions” (EMI) Nov 2025
- **2024 Inaugural Conference for Managing University Startups** Casablanca, Morocco
Presenter: “DomotiCool: AI-powered smart devices platform” (UH2C) July 2024
- **International Conference on Optimization and Learning (OLA 2024)** Dubrovnik, Croatia
Presenter: “Machine Learning Applications in Road Safety: A Comprehensive Review” May 2024

Schools, Workshops & Certifications

Winter/Summer & Spring Schools

- **OxML 2026 (Representation Learning & Generative AI)**, *University of Oxford, UK* Jul 2026
- **Responsible ML Winter School**, *Umeå University, Sweden* Mar 2025
- **Winter School: Next-Gen AI & Economic Applications**, *UM6P, Benguerir* Feb 2025
- **Spring School: Multimodal Foundation Models & GenAI**, *INPT, Rabat* Apr 2024
- **Winter School on Innovation and Valorization**, *Unite and Inoval360 (Remote)* Jan 2024

Workshops & Seminars

- **AI Technologies in Public Service: Needs and Challenges**, *Umeå University (Remote)* Apr 2025
- **IEOM Workshop (Six Sigma & Innovation and Entrepreneurship)**, *INPT, Rabat* Apr 2025
- **Wolfram Training: Mathematica & System Modeler**, *EMI, Rabat* Apr 2024
- **Workshop: "Initiation to Research: Academic Tools"**, *EMI, Rabat* Apr 2024
- **International Seminar: National Road Safety Strategies**, *Conrad Arzana, Rabat* Feb 2024

Technical Skills

- **AI & Machine Learning:** Deep Learning, Advanced Decision Trees, Explainable AI (XAI), AutoML, Time Series Forecasting, Large Language Models (LLMs), Prompt Engineering.
- **Programming & Data:** Python, SQL, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Dash, R.
- **Big Data & Tools:** Apache Hadoop, Hive, NiFi, MongoDB, HBase, MySQL.
- **Web & BI:** React, Flask, Power BI, Tableau.
- **Dev Environments:** Google Colab, Jupyter, VS Code, RStudio, Git/GitHub.
- **Operating Systems:** Linux, MacOS, Windows.
- **Research:** L^AT_EX, Zotero, Elicit, ResearchRabbit.

Languages

Arabic (Native), French (Advanced), English (Upper Intermediate).